# Chapter 10: Space Missions - Question Bank

## 1. Choose the correct option.

1. Due to _____, we can gather information about worldwide events sitting at home.
   - world wide web
   - internet
   - artificial satellite
   - natural satellite
2. The first person to step on the moon is _____.
   - Neil Armstrong
   - Rakesh Sharma
   - Kalpana Chawla
   - Sunita Williams
3. The first artificial satellite _____ was sent to space by Soviet Union in 1957.
   - Apollo
   - Chandrayaan
   - Sputnik
   - Luna 2
4. If a spacecraft is to be sent to travel to outer space, it must have a minimum velocity of _____.
   - $11.2\ \mathrm{km/s}$
   - $11.6\ \mathrm{km/s}$
   - $13.2\ \mathrm{km/s}$
   - $1.4\ \mathrm{m/s}$
5. A group of students from COEP Pune sent a small satellite _____ through ISRO in 2016.
   - Luna 6
   - Apollo 6
   - Swayam
   - Param
6. The astronomical object closest to us is the _____.
   - Moon
   - Mars
   - Saturn
   - Mercury

## 2. Find odd one out.

1. Find the odd one out and give its explanation:
   - Sputnik
   - Moon
   - Swayam
   - Chandrayaan

## 3. Right or Wrong sentence

1. The escape velocity on the moon is less than that on the earth.
2. India is the first country to discover the existence of water on the moon.
3. The function of a satellite launcher is based on Newton's second law of motion.

## 4. Write scientific reasons

1. The velocity of spacecraft at the earth's surface must be greater than the escape velocity of the earth.
2. Space debris can be harmful to artificial satellites.
3. Satellite launch vehicles are used to place satellites in their specific orbits.
4. The Launch vehicles are very costly.

## 5. Solve the following Questions

1. What is meant by space debris?
2. Why there is need to manage the debris?
3. Bring out the contribution of India's space missions.

4. Calculate the critical velocity of the satellite to be located at $35780\ \text{km}$ above the surface of earth.
5. What is Medium Earth Orbit?

# 6. Distinguish between

1. High Earth orbit - Medium Earth orbit.

# 7. Write short notes

1. Moon missions

# 8. Answer the Following

1. Observe the figure and write the answers:
    1. Name the outer orbit.
    2. Which satellites revolve in low earth orbits?
    3. Which various orbits are given in the figure?
    4. Give an example of a launch vehicle based on Newton's third law.
2. Explain why spacecraft take longer to reach the moon than light?
3. Write functions of Military satellite and Navigational satellite.
4. What is meant by an artificial satellite? How are the satellites classified based on their functions?
5. Solve the problem: If the mass of a planet is eight times the mass of the Earth and its radius is twice the radius of the Earth, what will be the escape velocity for that planet?
6. Explain: Escape velocity on the moon is less than escape velocity on the earth.
7. Complete the following table.

# 9. Answer the following questions (Section 1)

1. Note the relationship between the entries in all the three columns in the table and rewrite the table:

| Column-1 (Location) | Column-2 Height from the earth's surface (km) | Column-3 g (m/s2) |
|---|---|---|
| Earth's surface (average) | 8.8 | 0.225 |
| Mount Everest | 36.6 | 9.81 |
| Maximum height ever reached by manmade balloon | 400 | 9.8 |
| Orbit of a typical weather satellite | 35700 | 9.77 |
| Orbit of communication satellite | 0 | 8.7 |

# 10. Answer the following questions (Section 2)

1. Write the names of India's satellite series and launchers?
2. What is meant by satellite launch vehicles? Explain a satellite launch vehicle developed by ISRO with the help of a schematic diagram.
3. Read the following paragraphs and write the answers:Next to the moon, the astronomical object nearest to the earth is Mars. Many nations sent spacecrafts to Mars. Mars mission is difficult and almost half the missions were unsuccessful. However, ISRO's performance in this mission is remarkable and we all must be proud of it. The spacecraft 'Mangalyaan' made by ISRO using minimum expenses was launched in November, 2013 and was placed into orbit around Mars in September, 2014. It obtained very useful information about the surface of Mars and the atmosphere around it.
    1. Next to the moon, which is the nearest celestial object to the earth?
    2. When did Mangalyaan placed into orbit around Mars by ISRO?
    3. What important information did ISRO's Mangalyaan obtain?
    4. Explain that critical velocity does not depend on the mass of the satellite.

4. Explain the need and importance of space missions.